

WEEK-WISE LESSON PLAN - AWS & DevOps

Week-1 (Modules-1,2,3)

Day-1: Cloud Computing Fundamentals & Service Models

Day-2: AWS Global Infrastructure & Account Setup

Day-3: IAM – Users, Roles, MFA

Day-4: EC2 – Instance Types & Pricing

Day-5: Launch Windows & Linux EC2

Day-6: Security Groups, Key Pairs, Elastic IP

Week-2 (Modules-4,5,6)

Day-1: EBS – Volumes & Snapshots

Day-2: Custom AMI Creation

Day-3: VPC – CIDR & Custom VPC Setup

Day-4: Subnets & Route Tables

Day-5: Internet Gateway & NAT Gateway

Day-6: Bastion Server Setup

Week-3 (Modules-7,8,9,10,11)

Day-1: Elastic Load Balancer & Target Groups

Day-2: Auto Scaling Groups & Launch Templates

Day-3: S3 – Buckets & Object Management

Day-4: S3 Versioning & Lifecycle Policies

Day-5: S3 Static Website Hosting

Day-6: CloudFront Distribution

Week-4 (Modules-12,13,14,15,Some topics in 16)

Day-1: EFS Setup

Day-2: Route 53 – DNS & Hosted Zones

Day-3: RDS – DB Instances

Day-4: Multi-AZ & Read Replicas

Day-5: DynamoDB

Day-6: CloudWatch Monitoring

Week-5 (Modules-16,17)

Day-1: SNS, SQS, SES

Day-2: Lambda – Serverless Basics

Day-3: Lambda Permissions & Versions

Day-4: Linux File System & Basic Commands

Day-5: Linux Process & Memory Management

Day-6: Linux Package Management & Filters

Week-6 (Modules-17,Capstone Project-1,2)

Day-1: Service & systemctl Commands

Day-2: AWS Capstone Project-1

Day-3: ML Model Deployment on AWS

Day-4: Serverless Portfolio Project

Day-5: CI/CD Pipeline on AWS

Day-6: Containerized Microservice Deployment

Week-7 (Capstone Project-2, Major Project)

Day-1: Gen-AI Major Project on AWS

Day-2: Project Implementation

Day-3: Project Implementation

Day-4: Monitoring & Optimization

Day-5: Project Testing & Debugging

Day-6: Project Review & Assessment

Week-8

Day-1: Introduction to DevOps & SDLC

Day-2: Shell Scripting Basics

Day-3: Loops, Functions, File Handling

Day-4: Git – Version Control Basics

Day-5: Git Branching & Merging

Day-6: GitHub Remote Repository

Week-9

Day-1: Maven Build Tool & POM

Day-2: Jenkins Introduction

Day-3: Jenkins Jobs & Pipelines

Day-4: Jenkins Security & Plugins

Day-5: CI/CD Pipeline Setup

Day-6: Terraform Basics

Week-10

Day-1: Terraform EC2 Deployment

Day-2: Docker Installation

Day-3: Docker Images & Containers

Day-4: Docker Compose

Day-5: Docker Networking

Day-6: Dockerfile & Docker Hub

Week-11

Day-1: Kubernetes Architecture

Day-2: Pods, Services & Deployments

Day-3: ConfigMaps & High Availability

Day-4: Ansible Introduction

Day-5: Ansible Playbooks & Roles

Day-6: Prometheus & Grafana Monitoring

Week-12

Day-1: GitHub Actions / GitLab CI

Day-2: Serverless CI/CD

Day-3: AI DevOps Project-1

Day-4: AI Agent for Kubernetes

Day-5: End-to-End CI/CD Pipeline

Day-6: Jenkins + Docker + K8s Integration

Week-13

Day-1: Monitoring Setup

Day-2: Final Capstone Implementation

Day-3: Project Debugging

Day-4: Project Optimization

Day-5: Final Review

Day-6: Assessment & Certification Review